

Optics Letters

Broadband adiabatic power splitter on an etchless thin-film lithium niobate platform

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Received 22 April 2025; revised 21 May 2025; accepted 21 May 2025; posted 27 May 2025; published 16 June 2025

Lithium niobate on insulator (LNOI) is a promising platform for photonic integrated circuits (PICs), and 3-dB power splitters are the most essential building blocks in PICs. In this Letter, we propose and demonstrate a broadband, fabrication-tolerant 3-dB power splitter based on the photonic bound state in the continuum (BIC) and using a spatial adiabatic passage structure on a polymer-loaded LNOI platform. The simulation results show that the proposed device can achieve a splitting ratio of 50%:50% with an excellent operating bandwidth of 180 nm (from 1450 nm to 1630 nm). The fabricated device on the EpoCore-loaded LNOI platform features a power-splitting ratio of $\sim 50.13\%:49.87\%$ for both output ports within the 1530–1630 nm wavelength range. With its outstanding performance and simple fabrication process, the demonstrated 3-dB power splitter is essential for on-chip optical interconnect systems. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

<https://doi.org/10.1364/OL.565904>

Photonic integrated circuits (PICs) are key technology in optical communication systems, offering superior performance and low power consumption [1]. Thanks to advancements in material systems and the maturation of fabrication processes, it is now possible to integrate active and passive photonic devices on various platforms, including SiO_2 [2], Si_3N_4 [3], and lithium niobate on insulator (LNOI) [4,5]. LNOI stands out as the most promising candidate among these different platforms due to its ultrabroad optical transparency window (350–5200 nm), large electro-optic coefficients, and significant second-order optical nonlinearity [6]. The LNOI platform leverages the properties of lithium niobate to enable high-performance optical devices, including high-speed electro-optic modulators [7], broadband frequency combs [8], and polarization management devices [9]. Nonetheless, further enriching the repertoire of passive components is imperative for fostering the development of monolithic optoelectronic integrated circuits incorporating active elements.

3-dB power splitters serve as indispensable building blocks within PICs. Owing to their utility, they are often employed for light distribution or act as integral components in various devices, including optical switches [10], multiplexers [11], and

modulators [12]. Various types of 3-dB power splitters have been demonstrated on the LNOI platform, mainly containing directional couplers (DCs) [13], multimode interference (MMI) couplers [14], and Y-branches [15]. The power splitting of DCs based on the principles of coupled-mode theory will lead to pronounced dispersive behavior across an extensive spectral range. Despite their compact footprint and low loss, the power-splitting ratio of DCs is highly sensitive to dispersion, ultimately limiting their operational bandwidth. The MMI couplers, another widely adopted configuration, are characterized by a broad transmission bandwidth and substantial fabrication tolerance, yet they have a relatively large footprint. With their compact design and broad operating bandwidth, the Y-branches may still experience mode mismatch loss when junction angles are not sufficiently small. Moreover, the fabrication of these devices needs highly accurate dry etching with electron-beam lithography (EBL); inevitably, this results in the formation of slanted sidewalls in waveguides and thereby constrains the achievable minimum spacing between adjacent waveguides.

The etchless hybrid LNOI platform that deposits a material with a low refractive index on the LNOI platform, such as silicon nitride [16,17] and polymers [18,19], can greatly simplify the complexity of the fabrication process and reduce costs. Low-loss bound states in the continuum (BICs) modes can be effectively confined within the polymer–LNOI hybrid waveguide. These modes formed within the continuous spectrum demonstrate negligible radiation loss due to their inherent symmetry-protected characteristics [20]. Moreover, rather than relying on EBL, devices fabricated on the polymer–LNOI hybrid platform can be accomplished using the photolithography technique. To date, devices have been proposed on polymer–LNOI hybrid platforms, including the mode (de)multiplexer [18], acousto-optic modulator [19], and thermo-optic modulator [21].

In this Letter, we propose and demonstrate a broadband 3-dB power splitter based on spatial adiabatic passage (SAP) on an EpoCore-loaded LNOI platform. The SAP-based splitter effectively splits optical power with high fabrication tolerance and less power imbalance across an ultra-wide bandwidth. EpoCore is a positive photoresist that exhibits exceptional CMOS compatibility, making it an essential material platform for realizing PICs. The EpoCore-loaded LNOI waveguides maintain the superior material properties of LNOI while eliminating the need for direct LNOI

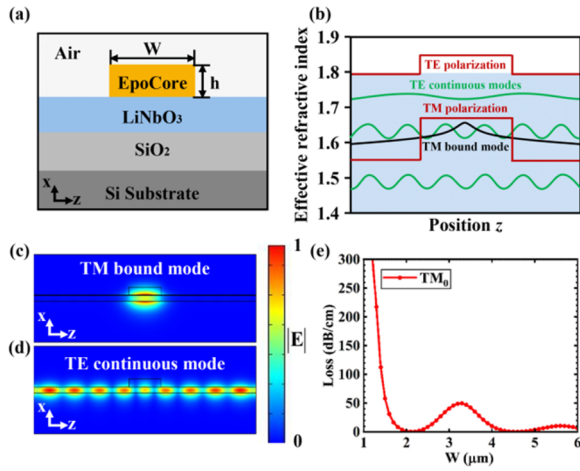


Fig. 1. (a) Cross-sectional view of the straight waveguide. (b) Effective refractive index distributions of the fundamental TE and TM modes. Electric field $|E|$ profiles of the (c) TM-bound mode and (d) the TE-continuous mode. (e) Propagation loss of the straight waveguide as a function of the waveguide width W at a wavelength of 1550 nm.

etching, thereby simplifying the fabrication process. The simulation results indicate that the splitting ratio between the two outputs is 50%:50% in the wavelength range of 1450–1630 nm. Experimental results demonstrate that the operation bandwidth spans the wavelength range from 1530 nm to 1630 nm, with a power-splitting ratio of approximately 50.13%:49.87%. With outstanding performance and an easy-to-implement manufacturing process, the demonstrated splitter can be applied in optical communication systems and PICs for efficient power splitting within the telecom wavelength range.

As schematically shown in Fig. 1(a), the cross section of a straight waveguide is constructed by patterning EpoCore on an x -cut LNOI substrate via photolithography and wet-etching processes. The thicknesses of the EpoCore, LNOI, and SiO_2 are 400 nm, 300 nm, and 4.7 μm , respectively. We employ the finite-element method to analyze the modes supported by the waveguide, where the refractive index of LNOI is set to $n_o = 2.211$ and $n_e = 2.138$, and the refractive index of EpoCore is 1.569 at the wavelength of 1550 nm. Figure 1(b) shows the effective refractive index distributions of the fundamental TE and TM modes at a wavelength of 1550 nm. The effective refractive index for the TM polarization is lower than that for the TE polarization, indicating that the TM-bound mode falls within the TE-continuous spectrum. The TM-polarized-bound mode coexists with the TE-polarized continuum, and their inevitable coupling leads to an optical loss in the TM-bound mode. However, the design of the structural parameters of the waveguide can effectively eliminate the interaction between the TM-bound and TE-continuous modes. This elimination significantly suppresses the energy dissipation of the bound mode, resulting in low leakage loss and consequently transforming the TM-bound mode into a bound state in the continuum [19]. Furthermore, Figs. 1(c) and 1(d) show the profile distribution ($|E|$) of the TM-bound and TE-continuous modes. Figure 1(e) illustrates the propagation losses of the TM_0 mode with different waveguide widths at a wavelength of 1550 nm, and it can be observed

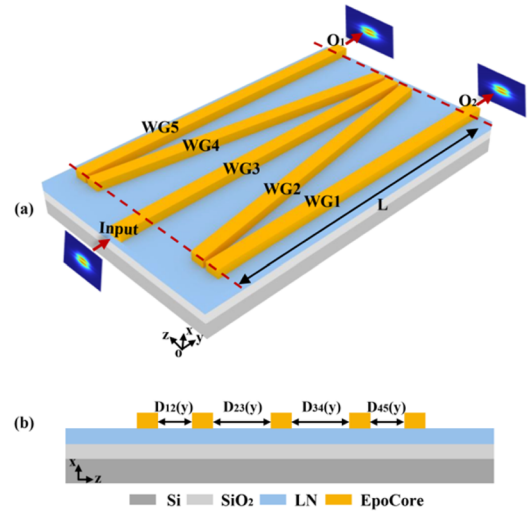


Fig. 2. (a) 3D schematic diagram and (b) cross-sectional view at $y = 0$ of the 5-waveguide SAP.

that the minimum propagation loss is 0.048 dB/cm when the waveguide width is 2.1 μm .

Figure 2(a) illustrates the 5-waveguide SAP, which consists of five straight waveguides with the same width, denoted as W , and the same length along the y axis, denoted as L . The origin of the y axis is positioned at the starting point of waveguide WG3. Waveguides WG2 and WG4 are flanked respectively by the outer waveguides WG1 and WG3, and WG3 and WG5, which are parallel to the y axis and separated by a distance enough to prevent coupling between the outer waveguides. Figure 2(b) shows the cross section of the 5-waveguide SAP at $y = 0$. $D_{12}(y)$, $D_{23}(y)$, $D_{34}(y)$, and $D_{45}(y)$ are the distances between waveguides WG1 and WG2, WG2 and WG3, WG3 and WG4, and WG4 and WG5, respectively, corresponding to coordinates on the y axis. Therefore, the distances between adjacent waveguides at $y = 0$ are denoted as $D_{12}(0)$ for WG1–WG2, $D_{23}(0)$ for WG2–WG3, $D_{34}(0)$ for WG3–WG4, and $D_{45}(0)$ for WG4–WG5, respectively. Waveguides WG2 and WG4 are placed mirror-symmetric about waveguide WG3, and designed such that $D_{12}(L/2) = D_{23}(L/2) = D_{34}(L/2) = D_{45}(L/2)$, ensuring that $D_{12}(0) = D_{23}(L) = D_{34}(L) = D_{45}(0)$ and $D_{12}(L) = D_{23}(0) = D_{34}(0) = D_{45}(L)$. The coupling coefficients between waveguides WG1 and WG2, WG2 and WG3, WG3 and WG4, WG4 and WG5, WG1 and WG3, and WG5 and WG3 can be expressed as [22]

$$\kappa_{12}(y) = \kappa_{45}(y) = \frac{\pi}{2L_{c,12}(y)} = \frac{\pi}{2L_{c0}e^{D_{12}(y)/r_0}}, \quad (1)$$

$$\kappa_{23}(y) = \kappa_{34}(y) = \frac{\pi}{2L_{c,23}(y)} = \frac{\pi}{2L_{c0}e^{D_{23}(y)/r_0}}, \quad (2)$$

$$\kappa_{13} = \kappa_{35} = \frac{\pi}{2L_{c,13}} = \frac{\pi}{2L_{c0}e^{D_{13}/r_0}}. \quad (3)$$

In Eq. (1), $L_{c,12}(y)$ is the coupling length for directional coupling between waveguides WG1 and WG2 (the same definition applies to $L_{c,23}(y)$ and $L_{c,13}(y)$). Parameters L_{c0} and r_0 are associated with the waveguide characteristic and operation wavelength, which is derived by measuring the coupling efficiency of a conventional directional coupler with varying lengths and spacings [22]. To ensure light adiabatic transport from waveguide WG3 to WG1 and WG5, $\kappa_{13}(0) = \kappa_{35}(0) \sim 0$ and

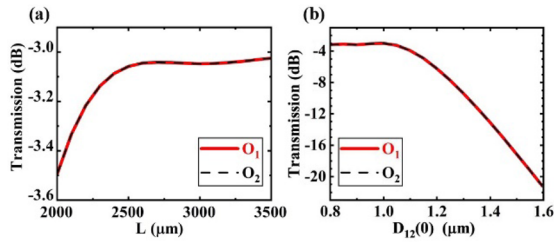


Fig. 3. Simulated transmission of outputs O_1 and O_2 as functions of (a) the length of the 5-waveguide SAP L and (b) the distance between WG1 and WG2 at $y = 0$, $D_{12}(0)$, at a wavelength of 1550 nm.

$\kappa_{12}(0) = \kappa_{45}(0) > \kappa_{23}(0) = \kappa_{34}(0)$ as a guideline for determining the coupling strength among the waveguides [22].

We utilized the three-dimensional finite-difference time-domain (3D-FDTD) method for numerical simulations to ensure an accurate and practical design. The grids in the cross section and the propagation direction of the waveguide are $0.1 \mu\text{m} \times 0.1 \mu\text{m}$ and $0.1 \mu\text{m}$, respectively. The length of the 5-waveguide SAP along the y axis L is designed to be sufficiently large to satisfy the adiabatic condition. To determine the optimal parameters, we varied L gradually from $2000 \mu\text{m}$ to $3500 \mu\text{m}$ at a wavelength of 1550 nm, and the relationship between transmission and L is illustrated in Fig. 3(a). The results reveal that an L value greater than $2700 \mu\text{m}$ yields the best performance, demonstrating that output ports O_1 and O_2 exhibit maximum transmission. Considering the footprint of the device, L is set to $2700 \mu\text{m}$. Considering the adiabatic condition and fabrication process, the distance between waveguides WG2 and WG3 at $y = 0$, $D_{23}(0)$, is set to $1.8 \mu\text{m}$. The distance $D_{12}(0)$ was adjusted to ascertain the precise parameter from $0.8 \mu\text{m}$ to $1.8 \mu\text{m}$. The variation in transmission for $D_{12}(0)$ at a wavelength of 1550 nm is depicted in Fig. 3(b), which indicates that the transmission gradually decreases as $D_{12}(0)$ increases. Considering the fabrication precision, larger device dimensions were adopted to ensure reliable performance. Therefore, $D_{12}(0)$ was ultimately selected to be $1.1 \mu\text{m}$.

Utilizing the optimized structural parameters, we further investigate the wavelength characteristics of the device by using the 3D-FDTD method. Figure 4(a) presents the corresponding electric field (E-field) distributions at wavelengths of 1450 nm, 1500 nm, 1550 nm, and 1600 nm. The results demonstrate efficient light coupling from the central waveguide (WG3) to the outer waveguides (WG1 and WG5) while showing negligible excitation in the intermediate waveguides (WG2 and WG4). This observation confirms the occurrence of adiabatic transport in broad bandwidth. The calculated transmission at output ports O_1 and O_2 as a function of wavelength is depicted in Fig. 4(b). A power-splitting ratio of approximately 50%:50% is achieved at ports O_1 and O_2 , with a maximum insertion loss of 0.27 dB across a bandwidth of 180 nm. The symmetrical configuration of the device results in a negligible power imbalance between the two output ports.

To ensure device reliability in practical fabrication, we simulated the transmission at both output ports, O_1 and O_2 , as functions of varying dimensions, including different widths and thicknesses. The detailed analyses of fabrication tolerances are presented in Figs. 4(c) and 4(d). Despite ± 100 nm variations in width (ΔW) and thickness (Δh), the worst-case insertion loss remains below 0.72 dB, and the splitting ratio between the two

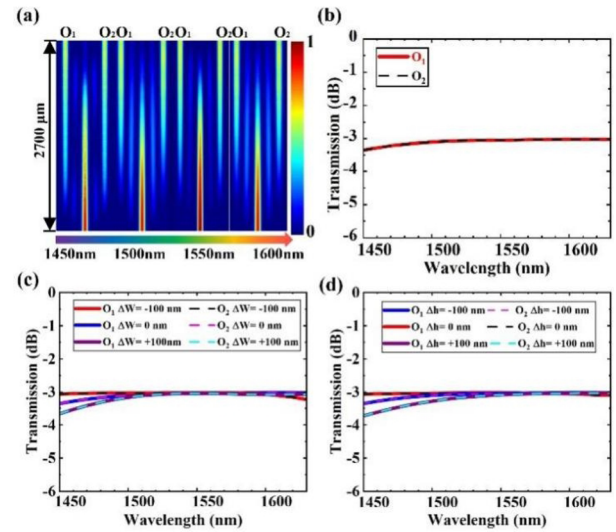


Fig. 4. (a) Intensity profiles of light propagation in the 5-waveguide SAP at wavelengths of 1450 nm, 1500 nm, 1550 nm and 1600 nm, respectively. (b) Simulated transmission spectra of the power splitter for the wavelength range spanning from 1450 nm to 1630 nm. Simulated transmission spectra as a function of (c) the waveguide width deviation, ΔW , and (d) the thickness of polymer deviation, Δh .

outputs is 50%:50% across the entire wavelength range from 1450 nm to 1630 nm. These simulation results highlight the robust fabrication tolerance of the proposed device.

The 5-waveguide SAP power splitter was fabricated using a microfabrication process, as depicted in Fig. 5(a). A 400-nm-thick EpoCore film was applied onto the x -cut LNOI wafer through spin coating, followed by prebaking at 40°C for 5 min and then at 90°C for 30 min to eliminate the solvent. Standard contact photolithography, employing a mask, was used to pattern the waveguides onto the EpoCore film. Subsequently, the film was exposed to ultraviolet (UV) light from a mercury discharge lamp for 10 s. After that, the sample was wet-etched in the developer for 3 s to remove the unexposed EpoCore material. The sample was then rinsed in isopropyl alcohol to remove residual adhesive, followed by a deionized water wash to eliminate any remaining reaction solution. Subsequently, it was hard-baked at 120°C for 40 min to enhance material cross-linking. Figures 5(b)–5(e) exhibit the top views and enlarged views of the fabricated device at its input, middle, and output sections, as observed through an optical microscope. The dimensions closely align with the intended design, which is beneficial for minimizing transmission losses and enhancing the overall performance of the device.

To evaluate the power-splitting functionality of the fabricated 5-waveguide SAP power splitter, light from a tunable semiconductor laser was coupled into the device via a tapered fiber. Simultaneously, the polarization state of the input light was controlled using a polarization controller. The near-field output spot of the device was captured using a charge-coupled device (CCD) camera. At the output port, the optical signal was coupled into a receiving optical fiber, which was connected to an optical powermeter (Newport 2832-C) for measuring the output power. Figure 6 illustrates the power-splitting ratio across the wavelength range from 1530 nm to 1630 nm, currently limited

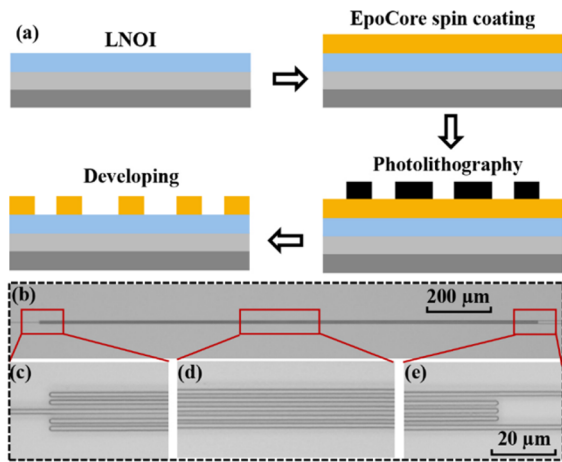


Fig. 5. (a) Steps for fabricating the device. (b) Top view and (c)–(e) enlarged view of a typical fabricated 5-waveguide SAP.

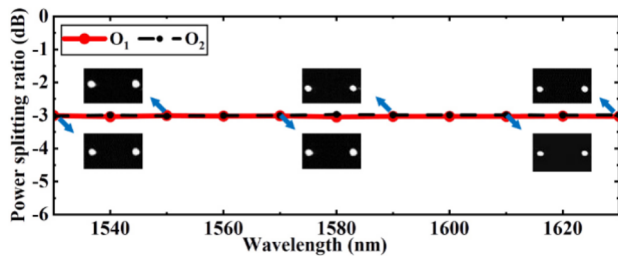


Fig. 6. Normalized power-splitting ratio at the wavelength range from 1530 nm to 1630 nm. The insets show the near-field images at O₁ and O₂ ports at wavelengths of 1530 nm, 1550 nm, 1570 nm, 1590 nm, 1610 nm, and 1630 nm, respectively.

by our test system. At the wavelength of 1550 nm, the normalized power-splitting ratio was measured to be 50.13%:49.87%, aligning well with the simulation results. Additionally, the output spatial images of the device were taken at wavelengths of 1530 nm, 1550 nm, 1570 nm, 1590 nm, 1610 nm, and 1630 nm, respectively. Our experimental measurements indicate that the device operates excellently within a bandwidth of 100 nm.

In conclusion, we propose and demonstrate a high-performance 3-dB power splitter on an EpoCore-loaded LNOI platform, featuring broadband operation, low imbalance, and large fabrication tolerance. By optimizing the dimensions of

SAP, we simultaneously achieve a power-splitting ratio of approximately 50%:50% and a significant fabrication tolerance of ± 100 nm. The measured operation bandwidth of the device spans 100 nm, from 1530 nm to 1630 nm. At a wavelength of 1550 nm, the power-splitting ratio is nearly 50.13%:49.87%. We believe that this broadband and fabrication-tolerant device presents a promising design approach and will play a significant role in future communication systems.

Funding. National Key Research and Development Program of China (2024YFE0211800); Jilin Province Development and Reform Commission (2023C030-7).

Disclosures. The authors declare no conflicts of interest.

Data availability. The data underlying the results presented in this Letter are not publicly available at this time but may be obtained from the authors upon reasonable request.

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